Polar Coordinates and Conversions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Converting points**

Convert each polar point (r, θ) to rectangular coordinates (exact values):

a $\left(4, \frac{\pi}{3}\right)$ $(2, 2\sqrt{3})$

b $\left(2, \frac{3\pi}{4}\right)$ $(-\sqrt{2}, \sqrt{2})$

(a) $(2, 2\sqrt{3})$; (b) $(-\sqrt{2}, \sqrt{2})$

2 **[Foundational]** • **Target: Converting points**

Convert each rectangular point to polar coordinates with $r \geq 0$ and $0 \leq \theta < 2\pi$:

a $(-3, 3)$ $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$

b $(0, -5)$ $\left(5, \frac{3\pi}{2}\right)$

(a) $\left(3\sqrt{2}, \frac{3\pi}{4}\right)$; (b) $\left(5, \frac{3\pi}{2}\right)$

3 **[Foundational]** • **Target: Converting equations**

Convert the polar equation $r = 6\cos\theta$ to rectangular form and identify the curve.

Multiply by r: $r^2 = 6r\cos\theta$, so $x^2 + y^2 = 6x$, i.e. $(x - 3)^2 + y^2 = 9$: a circle with center $(3, 0)$ and radius 3.

4 **[Foundational]** • **Target: Converting equations**

Convert each rectangular equation to polar form:

a $x^2 + y^2 = 16$ $r = 4$

b $y = x$ $\theta = \frac{\pi}{4}$

(a) $r = 4$; (b) $\theta = \frac{\pi}{4}$

5 **[Foundational]** • **Target: Converting equations**

Convert the polar equation $r = \frac{2}{1 - \sin\theta}$ to rectangular form.

$r - r\sin\theta = 2$, so $r = y + 2$. Squaring: $x^2 + y^2 = (y + 2)^2 = y^2 + 4y + 4$, so $x^2 = 4y + 4$ — a parabola.

6 **[Foundational]** • **Target: Multiple representations**

Give two other pairs (r, θ) that represent the same point as $\left(3, \frac{\pi}{4}\right)$: one with $r > 0$ and θ outside $[0, 2\pi)$, and one with $r < 0$.

Adding a full rotation: $\left(3, \frac{\pi}{4} + 2\pi\right) = \left(3, \frac{9\pi}{4}\right)$. Using negative r (add π to the angle): $\left(-3, \frac{\pi}{4} + \pi\right) = \left(-3, \frac{5\pi}{4}\right)$.

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7 [Foundational] • Target: Multiple representations

Plot-reasoning: explain why the polar point $(-2, \frac{\pi}{2})$ corresponds to the same location as $(2, \frac{3\pi}{2})$.

A negative r means the point lies in the direction opposite to θ. Pointing at angle $\frac{\pi}{2}$ (straight up) with $r = -2$ places the point 2 units in the opposite direction, straight down — the same location as $r = 2$ at angle $\frac{3\pi}{2}$ (also straight down).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distance between polar points

Convert $(4, \frac{\pi}{6})$ and $(4, \frac{2\pi}{3})$ to rectangular coordinates, then find the distance between them, to two decimal places.

$(4, \frac{\pi}{6}) \to (2\sqrt{3}, 2)$; $(4, \frac{2\pi}{3}) \to (-2, 2\sqrt{3})$. Distance $= \sqrt{(2\sqrt{3} + 2)^2 + (2 - 2\sqrt{3})^2} \approx \sqrt{29.86 + 2.14} = \sqrt{32} \approx 5.66$.

9 [Intermediate] • Target: Further conversion practice

Convert each polar point to rectangular coordinates (exact values):

a $(6, \pi)$ $(-6, 0)$

b $(5, \frac{5\pi}{6})$ $\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$

(a) $(-6, 0)$; (b) $\left(-\frac{5\sqrt{3}}{2}, \frac{5}{2}\right)$

10 [Intermediate] • Target: Further conversion practice

Convert the polar equation $r^2 = 4\sin 2\theta$ to note its symmetry, and identify it as a standard curve type.

This is a lemniscate (figure-eight curve). Rewriting $r^2 = 4\sin 2\theta = 8\sin\theta\cos\theta$, and using $r^2 = x^2 + y^2$, $r\cos\theta = x$, $r\sin\theta = y$: multiplying both sides by r^2 gives $(x^2 + y^2)^2 = 8xy$.

11 [Intermediate] • Target: Why polar coordinates are not unique

Explain why, unlike rectangular coordinates, a single point can be represented by infinitely many polar coordinate pairs.

Adding any multiple of 2π to θ returns to the same direction, and reversing the sign of r while adding π to θ also gives the same point — so every point has infinitely many valid (r, θ) representations, unlike the one-to-one rectangular system.

12 [Intermediate] • Target: Why polar coordinates are not unique

The pole (origin) is a special case in polar coordinates. Explain why.

The pole corresponds to $r = 0$ for ANY value of θ — every angle gives the same point (the origin) when $r = 0$, so the pole has no single well-defined angle.

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13 [Intermediate] • Target: Converting more complex equations

Convert the rectangular equation $x^2 - y^2 = 4$ (a hyperbola) to polar form.

$r^2\cos^2\theta - r^2\sin^2\theta = 4 \Rightarrow r^2(\cos^2\theta - \sin^2\theta) = 4 \Rightarrow r^2\cos(2\theta) = 4.$

14 [Intermediate] • Target: More point conversions with negative r

Convert the polar point $(-4, \frac{\pi}{6})$ (negative r) to rectangular coordinates.

$x = r\cos\theta = -4\cos\frac{\pi}{6} = -4 \cdot \frac{\sqrt{3}}{2} = -2\sqrt{3}$. $y = r\sin\theta = -4\sin\frac{\pi}{6} = -4 \cdot \frac{1}{2} = -2$. Point: $(-2\sqrt{3}, -2)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: More point conversions with negative r • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Convert the rectangular point $(-4, -4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

$r = \sqrt{16 + 16} = 4\sqrt{2}$. Both coordinates negative places this in QIII, reference angle $\frac{\pi}{4}$: $\theta = \pi + \frac{\pi}{4} = \frac{5\pi}{4}$.

16 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 2.02 Radian Measure and Arc Length

The Archimedean spiral $r = \theta$ has no simple rectangular equivalent, but explain qualitatively why it spirals outward as θ increases.

Since $r = \theta$, the radius grows in direct proportion to the angle swept — each additional full rotation (2π radians) pushes r out by 2π more units, producing evenly-spaced spiral arms rather than a closed curve.

17 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A radar station models detected objects using polar coordinates (r, θ) where r is distance in km and θ is bearing angle. An aircraft is detected at $(r, \theta) = (45, \frac{5\pi}{6})$. Convert this to rectangular coordinates (kilometers east and north of the station), to two decimal places.

$x = 45\cos\frac{5\pi}{6} = 45\left(-\frac{\sqrt{3}}{2}\right) \approx -38.97$ km. $y = 45\sin\frac{5\pi}{6} = 45\left(\frac{1}{2}\right) = 22.50$ km. The aircraft is about 38.97 km west and 22.50 km north of the station.

18 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

A second aircraft is detected at rectangular coordinates $(30, -40)$ relative to the same radar station. Convert to polar coordinates (r, θ) with $r > 0$ and $0 \le \theta < 2\pi$, to two decimal places.

$r = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$ km. Reference angle $= \tan^{-1}\left(\frac{40}{30}\right) \approx 53.13° \approx 0.93$ rad. Since $x > 0, y < 0$ (QIV): $\theta \approx 2\pi - 0.93 \approx 5.35$ rad.

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19 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Convert the rectangular equation of a vertical line, $x = 5$, to polar form, and explain why the result is not simply a constant r.

$r\cos\theta = 5 \Rightarrow r = \frac{5}{\cos\theta} = 5\sec\theta$. Unlike a circle centered at the pole (constant r), a vertical line's distance from the pole varies with direction θ, so r must depend on θ through the secant function.

20 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Convert the polar equation $r = 6$ to rectangular form, and explain in one sentence why this equation contains no θ term.

$x^2 + y^2 = 36$ (squaring both sides of $r = 6$). There is no θ term because every direction from the pole at distance 6 satisfies the equation equally — the curve is a full circle, independent of angle.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Plotting with Positive and Negative Radii

How is a point with negative radius $(-r, \theta)$ plotted in polar coordinates?

a. Plot along the opposite ray directly across the pole (angle $\theta + \pi$) ✓

b. Plot in the negative y direction

c. Reflect across the y-axis

d. The point is undefined

A negative radius $-r$ means moving a distance r backward along the line passing through angle θ, which lands at $(r, \theta + \pi)$.

22 [Foundational] • Target: Plotting with Positive and Negative Radii

Which polar coordinate represents the same point as $(3, \pi/3)$?

a. $(-3, 4\pi/3)$ ✓

b. $(-3, 2\pi/3)$

c. $(3, 4\pi/3)$

d. $(-3, -\pi/3)$

Adding π to the angle and negating the radius gives $(-3, \pi/3 + \pi) = (-3, 4\pi/3)$.

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23 [Foundational] • Target: Plotting with Positive and Negative Radii

In which quadrant does the polar point $(4, 7\pi/6)$ lie?

a. Quadrant III ✓

b. Quadrant II

c. Quadrant IV

d. Quadrant I

Angle $7\pi/6 = 210^\circ$ lies in Quadrant III. With positive radius $r = 4$, the point is in Quadrant III.

24 [Foundational] • Target: Converting Polar to Cartesian Coordinates

State the formulas to convert polar coordinates (r, θ) to Cartesian coordinates (x, y).

a. $x = r\cos\theta, y = r\sin\theta$ ✓

b. $x = r\sin\theta, y = r\cos\theta$

c. $x = r^2\cos\theta, y = r^2\sin\theta$

d. $x = \frac{\cos\theta}{r}, y = \frac{\sin\theta}{r}$

By standard right-triangle trigonometry, $x = r\cos\theta$ and $y = r\sin\theta$.

25 [Foundational] • Target: Converting Polar to Cartesian Coordinates

Convert the polar point $(6, 2\pi/3)$ to exact Cartesian coordinates.

a. $(-3, 3\sqrt{3})$ ✓

b. $(3, -3\sqrt{3})$

c. $(-3\sqrt{3}, 3)$

d. $(-3, -3\sqrt{3})$

$x = 6\cos(2\pi/3) = 6(-1/2) = -3$. $y = 6\sin(2\pi/3) = 6(\sqrt{3}/2) = 3\sqrt{3}$.

26 [Foundational] • Target: Converting Polar to Cartesian Coordinates

Convert $(4, -\pi/4)$ to Cartesian coordinates.

a. $(2\sqrt{2}, -2\sqrt{2})$ ✓

b. $(-2\sqrt{2}, 2\sqrt{2})$

c. $(2\sqrt{2}, 2\sqrt{2})$

d. $(4, -4)$

$x = 4\cos(-\pi/4) = 4(\sqrt{2}/2) = 2\sqrt{2}$. $y = 4\sin(-\pi/4) = 4(-\sqrt{2}/2) = -2\sqrt{2}$.

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27 [Foundational] • Target: Converting Cartesian to Polar Coordinates

Convert the Cartesian point $(-2, 2)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(2\sqrt{2}, 3\pi/4)$ ✓

b. $(2\sqrt{2}, 5\pi/4)$

c. $(4, 3\pi/4)$

d. $(2\sqrt{2}, \pi/4)$

$r = \sqrt{(-2)^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$. In Quadrant II, $\theta = 3\pi/4$.

28 [Foundational] • Target: Converting Cartesian to Polar Coordinates

Convert $(0, -5)$ to polar coordinates.

a. $(5, 3\pi/2)$ ✓

b. $(5, \pi/2)$

c. $(-5, 0)$

d. $(5, \pi)$

Distance $r = 5$. The point lies on the negative y-axis, which is angle $\theta = 3\pi/2$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Converting Cartesian to Polar Coordinates

Convert $(3, -3\sqrt{3})$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $(6, 5\pi/3)$ ✓

b. $(6, 4\pi/3)$

c. $(6, 2\pi/3)$

d. $(12, 5\pi/3)$

$r = \sqrt{9 + 27} = \sqrt{36} = 6$. In Quadrant IV with $\tan\theta = -\sqrt{3}$, $\theta = 2\pi - \pi/3 = 5\pi/3$.

30 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

Convert the polar equation $r = 4\cos\theta$ to Cartesian form.

a. $(x-2)^2 + y^2 = 4$ ✓

b. $x^2 + y^2 = 4$

c. $x^2 + (y-2)^2 = 4$

d. $x = 4$

Multiply by r: $r^2 = 4r\cos\theta \Rightarrow x^2 + y^2 = 4x \Rightarrow (x-2)^2 + y^2 = 4$.

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31 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

Convert $r = 5\csc\theta$ to Cartesian form.

a. $y = 5$ ✓

b. $x = 5$

c. $x^2 + y^2 = 25$

d. $y = 5x$

$r = \frac{5}{\sin\theta} \Rightarrow r\sin\theta = 5 \Rightarrow y = 5$.

32 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

Convert the line $y = 3x$ to a polar equation.

a. $\theta = \arctan(3)$ ✓

b. $r = 3$

c. $r = 3\cos\theta$

d. $\theta = 3$

Substitute $x = r\cos\theta, y = r\sin\theta$: $r\sin\theta = 3r\cos\theta \Rightarrow \tan\theta = 3 \Rightarrow \theta = \arctan(3)$.

33 [Intermediate] • Target: Converting Cartesian to Polar Coordinates

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. **5** ✓

b. 7

c. 25

d. 16

The complex number can be written as $z = 3 + 4i$. The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. In this case, $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

34 [Intermediate] • Target: Converting Equations Between Polar and Cartesian

What is the Cartesian equation of the polar curve $r = 10$?

a. $x^2 - y^2 = 100$

b. $x^2 + y^2 = 100$ ✓

c. x+y=100

d. $x^2 + y^2 = 10$

Applying Polar Coordinates principles, Squaring $r = 10$: $r^2 = x^2 + y^2 = 100$. Therefore, the verified correct option is $x^2 + y^2 = 100$.

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35 [Intermediate] • Target: Converting points

What is the Cartesian equation of the polar curve $r = 11$?

a. $x^2 + y^2 = 121$ ✓

b. $x^2 + y^2 = 11$

c. $x^2 - y^2 = 121$

d. x+y=121

Applying Polar Coordinates principles, Squaring $r = 11$: $r^2 = x^2 + y^2 = 121$. Therefore, the verified correct option is $x^2 + y^2 = 121$.

36 [Intermediate] • Target: Converting equations

What is the Cartesian equation of the polar curve $r = 12$?

a. $x^2 + y^2 = 12$

b. $x^2 - y^2 = 144$

c. x+y=144

d. $x^2 + y^2 = 144$ ✓

Applying Polar Coordinates principles, Squaring $r = 12$: $r^2 = x^2 + y^2 = 144$. Therefore, the verified correct option is $x^2 + y^2 = 144$.

37 [Intermediate] • Target: Multiple representations

What is the Cartesian equation of the polar curve $r = 13$?

a. $x^2 + y^2 = 13$

b. $x^2 - y^2 = 169$

c. x+y=169

d. $x^2 + y^2 = 169$ ✓

Applying Polar Coordinates principles, Squaring $r = 13$: $r^2 = x^2 + y^2 = 169$. Therefore, the verified correct option is $x^2 + y^2 = 169$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Distance between polar points • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

What is the Cartesian equation of the polar curve $r = 14$?

a. $x^2 + y^2 = 196$ ✓

b. x+y=196

c. $x^2 - y^2 = 196$

d. $x^2 + y^2 = 14$

Applying Polar Coordinates principles, Squaring $r = 14$: $r^2 = x^2 + y^2 = 196$. Therefore, the verified correct option is $x^2 + y^2 = 196$.

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39 [Challenging] • Target: Further conversion practice • Cross-Lesson: 3.01 The Unit Circle and Exact Values

What is the Cartesian equation of the polar curve $r = 15$?

a. $x^2 + y^2 = 225$ ✓

b. $x^2 + y^2 = 15$

c. x+y=225

d. $x^2 - y^2 = 225$

Applying Polar Coordinates principles, Squaring $r = 15$: $r^2 = x^2 + y^2 = 225$. Therefore, the verified correct option is $x^2 + y^2 = 225$.

40 [Challenging] • Target: Why polar coordinates are not unique • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

What is the Cartesian equation of the polar curve $r = 16$?

a. $x^2 + y^2 = 16$

b. x+y=256

c. $x^2 - y^2 = 256$

d. $x^2 + y^2 = 256$ ✓

Applying Polar Coordinates principles, Squaring $r = 16$: $r^2 = x^2 + y^2 = 256$. Therefore, the verified correct option is $x^2 + y^2 = 256$.

41 [Challenging] • Target: Converting more complex equations • Cross-Lesson: 3.03 Sum and Difference Formulas

What is the Cartesian equation of the polar curve $r = 17$?

a. x+y=289

b. $x^2 + y^2 = 289$ ✓

c. $x^2 - y^2 = 289$

d. $x^2 + y^2 = 17$

Applying Polar Coordinates principles, Squaring $r = 17$: $r^2 = x^2 + y^2 = 289$. Therefore, the verified correct option is $x^2 + y^2 = 289$.

42 [Challenging] • Target: More point conversions with negative r • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the Cartesian equation of the polar curve $r = 18$?

a. x+y=324

b. $x^2 - y^2 = 324$

c. $x^2 + y^2 = 324$ ✓

d. $x^2 + y^2 = 18$

Applying Polar Coordinates principles, Squaring $r = 18$: $r^2 = x^2 + y^2 = 324$. Therefore, the verified correct option is $x^2 + y^2 = 324$.

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43 [Challenging] • Target: Converting a spiral equation • Cross-Lesson: 4.01 Circles and Parabolas

What is the Cartesian equation of the polar curve $r = 19$?

a. x+y=361

b. $x^2 - y^2 = 361$

c. $x^2 + y^2 = 361$ ✓

d. $x^2 + y^2 = 19$

Applying Polar Coordinates principles, Squaring $r = 19$: $r^2 = x^2 + y^2 = 361$. Therefore, the verified correct option is $x^2 + y^2 = 361$.

44 [Challenging] • Target: Converting a real-world radar equation • Cross-Lesson: 4.02 Ellipses

What is the Cartesian equation of the polar curve $r = 20$?

a. $x^2 - y^2 = 400$

b. x+y=400

c. $x^2 + y^2 = 400$ ✓

d. $x^2 + y^2 = 20$

Applying Polar Coordinates principles, Squaring $r = 20$: $r^2 = x^2 + y^2 = 400$. Therefore, the verified correct option is $x^2 + y^2 = 400$.

45 [Challenging] • Target: Plotting with Positive and Negative Radii • Cross-Lesson: 4.03 Hyperbolas

What is the Cartesian equation of the polar curve $r = 4$?

a. $x^2 - y^2 = 16$

b. x+y=16

c. $x^2 + y^2 = 4$

d. $x^2 + y^2 = 16$ ✓

Applying Polar Coordinates principles, Squaring $r = 4$: $r^2 = x^2 + y^2 = 16$. Therefore, the verified correct option is $x^2 + y^2 = 16$.

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